# Python-Seaborn

# Objective

- Basics of Seaborn
- Installation
- Seaborn Figure Styles
- Types Of Plots
- Color Palette
- Summary

# Basics of Seaborn

Seaborn is a library mostly used for statistical plotting in Python. It is built on top of Matplotlib and provides beautiful default styles and color palettes to make statistical plots more attractive.

**Important Features of Seaborn**

1. Built in themes for styling matplotlib graphics.
2. Visualizing univariate and bivariate data.
3. Fitting in and visualizing linear regression models.
4. Plotting statistical time series data.
5. Seaborn works well with NumPy and Pandas data structures.
6. It comes with built in themes for styling Matplotlib graphics.

# Installation

- use following command to install Seaborn
  pip install seaborn

- To check Seaborn version
  import seaborn
  print(seaborn.__version__)

**If you are using Anaconda**

- Start Navigator.
- In Environments  tab checked for installed packages.



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# Seaborn Figure Styles

The interface for manipulating the styles is set_style(). Using this function you can set the theme of the plot. As per the latest updated version, below are the five themes available.

1. Darkgrid
2. Whitegrid
3. Dark
4. White
5. Ticks



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# Types Of Plots

1. **Relational plots**: This plot is used to understand the relation between two variables.
2. **Categorical plots**: This plot deals with categorical variables and how they can be visualized.
3. **Distribution plots**: This plot is used for examining univariate and bivariate distributions
4. **Regression plots**: The regression plots in seaborn are primarily intended to add a visual guide that helps to emphasize patterns in a dataset during exploratory data analyses.
5. **Matrix plots:** A matrix plot is an array of scatterplots.
6. **Multi-plot grids:** It is an useful approach is to draw multiple instances of the same plot on different subsets of the dataset.

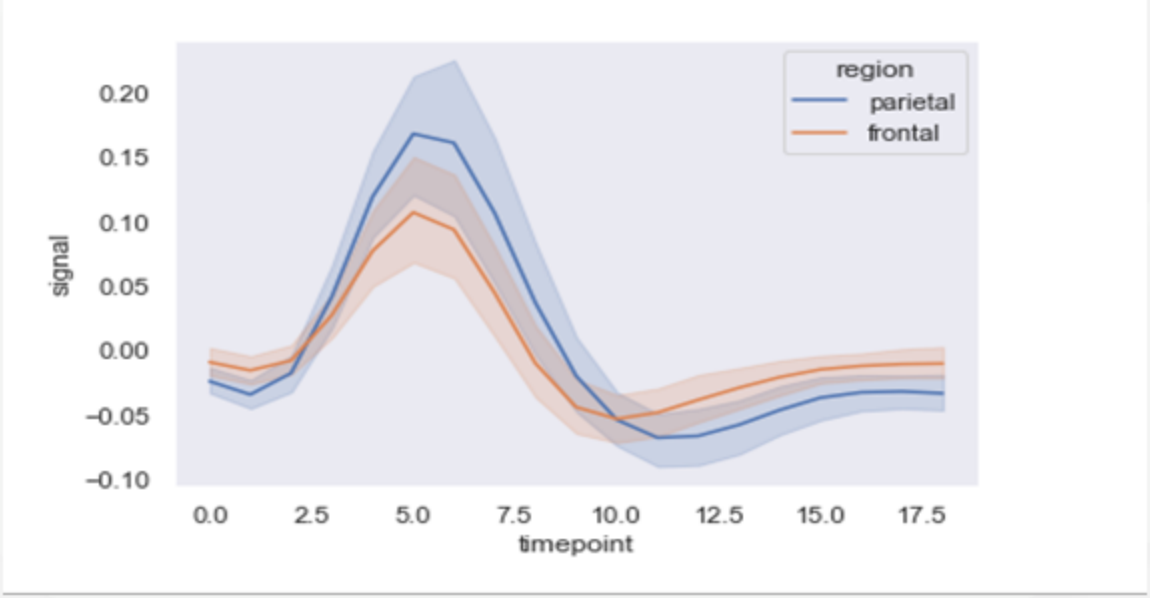
# Color Palette

- Colormaps are used to visualize plots effectively and easily. One might use different sorts of colormaps for different kinds of plots.
- color_palette() method is used to give colors to the plot.
- Another function palplot() is used to deal with the color palettes and plots the color palette as a horizontal array.

```
#importing packages
import seaborn as sns
import matplotlib.pyplot as plt
# current colot palette
palette = sns.color_palette()
# plots the color palette as a horizontal array
sns.palplot(palette)
plt.show()
```

# lineplot

```
#lineplot
import seaborn as sns
import matplotlib.pyplot as plt
sns.set(style="dark")
fm= sns.load_dataset("fmri")
sns.lineplot(x="timepoint",y="signal",hue="region",data=fm)
```



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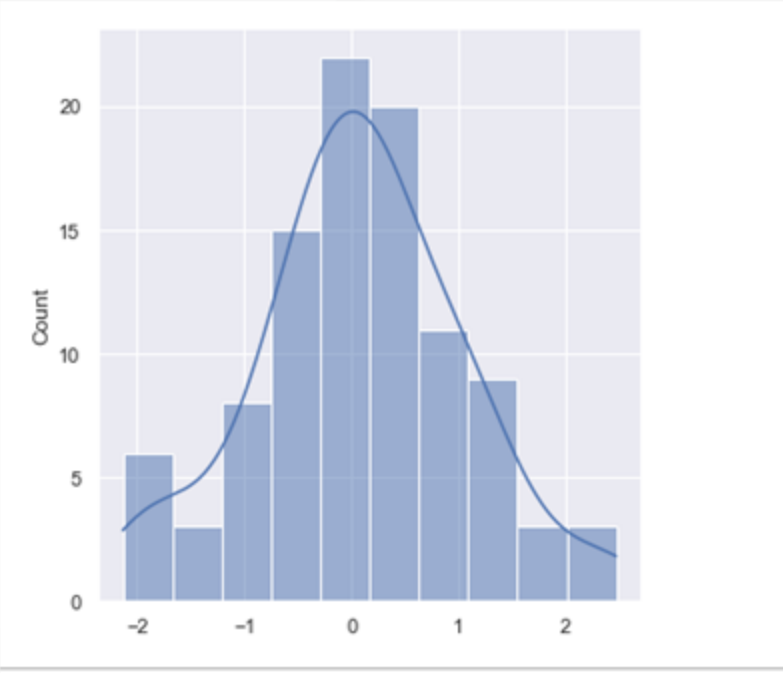
# Dis plot

```
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
sns.set(style='darkgrid') #dark,ticks,whitegrid,darkgrid
rs = np.random.RandomState(10)
d= rs.normal(size=100)
sns.displot(d,kde=True,color="b")  #b,r,m
```



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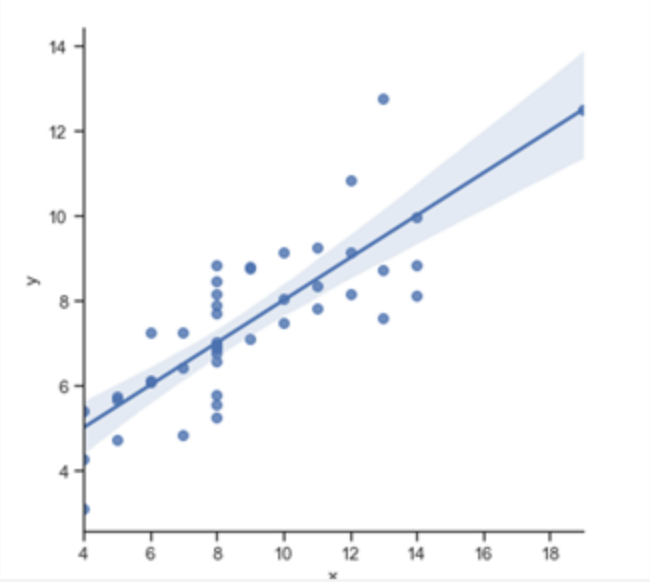
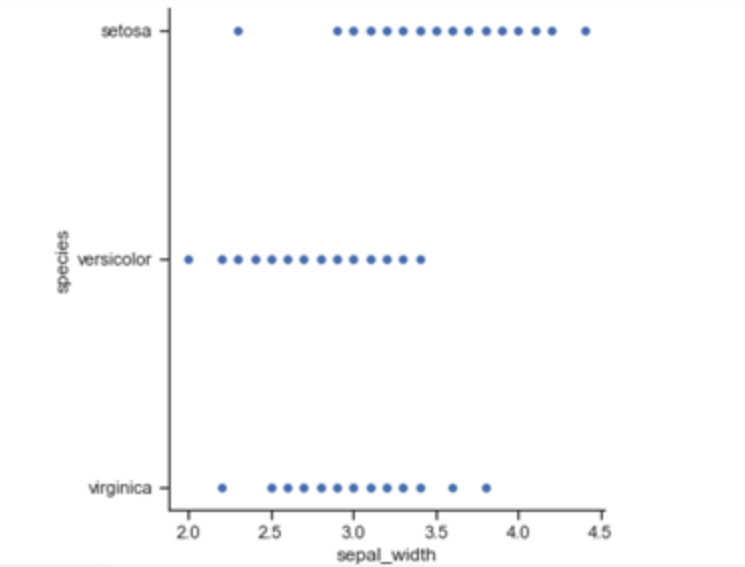
# lmplot

```
#lmplot
import seaborn as sns
import matplotlib.pyplot as plt
sns.set(style="ticks")
fm= sns.load_dataset("anscombe")
sns.lmplot(x="x",y="y",data=fm)
```



# Relational plots

```
#relational plots
import seaborn as sns
import matplotlib.pyplot as plt
fm= sns.load_dataset("iris")
sns.relplot(x="sepal_width",y="species",data=fm)
plt.show()
```



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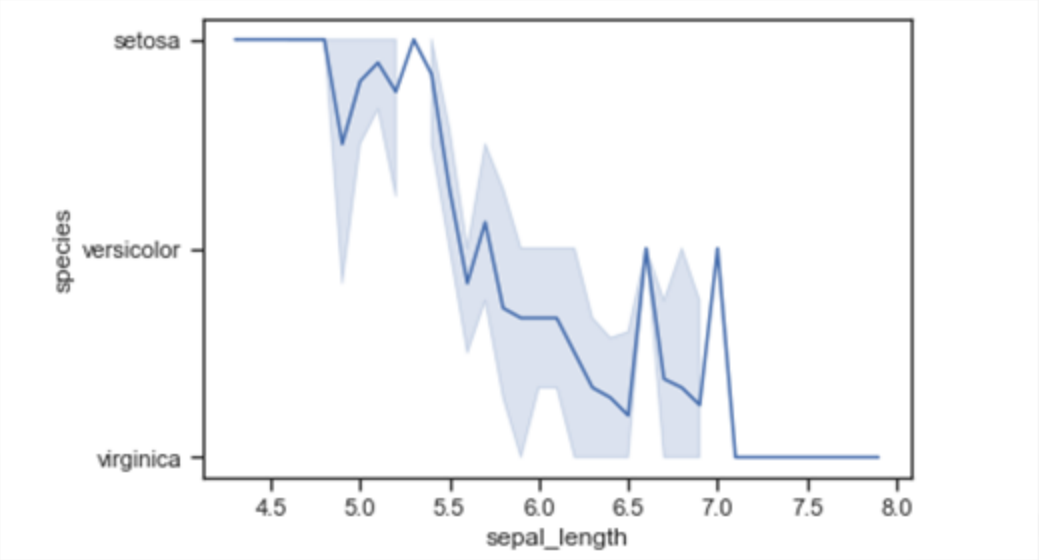
# Line plot

```
import seaborn as sns
import matplotlib.pyplot as plt
fm= sns.load_dataset("iris")
sns.lineplot(x="sepal_length",y="species",data=fm)
plt.show()
```



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# Thank You